

14) Use quantifiers and predicates with more than one variable to express these statements.

a) There is a student in this class who can speak Hindi.

$Q(x, y)$: Student x can speak language y .

Domain for x : All students in the class

Domain for y : All languages in the world

$\exists x Q(x, \text{Hindi})$

b) Every student in this class plays some sport

~~$Q(x, y)$~~ $P(x, y)$: Student x plays sport y .

Domain for x : All students in the class

Domain for y : All sports in the world

$\forall x \exists y P(x, y)$

c) Some student in this class has visited Alaska, but has not visited Hawaii.

$V(x, y)$: Student x has visited the place y .

Domain for x : All students in the class

Domain for y : All the places in the world

$\exists x (V(x, \text{Alaska}) \wedge \neg V(x, \text{Hawaii}))$

d) All students in this class have learned at least one programming language

$L(x, y)$: Student x has learnt programming language y .

Domain for x : All students in the class

Domain for y : All the programming languages in the world

$\forall x \exists y L(x, y)$

e) There is a student in this class who has taken every course offered by one of the departments in this school

$T(x, y, z)$: Student x has taken course z from department y

Domain for x : All students in the class

Domain for y : All the departments in the school